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 Studierichting: Informatica
 Oefeningenreeks 6G nr. 2.3

$$f(x) = |\sin x| \text{ op } [-\pi, \pi]$$

$$f(x) \text{ is even} \Rightarrow b_n = 0$$

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} |\sin x| dx = \frac{2}{\pi} \int_0^{\pi} \sin x dx \\ &= \frac{2}{\pi} [-\cos x]_0^{\pi} = \frac{2}{\pi} (1 - (-1)) = \frac{4}{\pi} \\ a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} |\sin x| \cos(nx) dx = \frac{2}{\pi} \int_0^{\pi} \sin x \cos(nx) dx \end{aligned}$$

$\Rightarrow$ Simpson:

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_0^{\pi} (\sin(1+n)x + \sin(1-n)x) dx \\ &= \frac{1}{\pi} \left[-\frac{\cos(1+n)x}{1+n} - \frac{\cos(1-n)x}{1-n} \right]_0^{\pi} \\ &= \frac{1}{\pi} \left(\frac{-\cos(1+n)\pi + 1}{1+n} + \frac{-\cos(1-n)\pi + 1}{1-n} \right) \\ \cos(1 \pm n)\pi &= -\cos(n\pi) = -(-1)^n \\ \Rightarrow a_n &= \frac{1}{\pi} \left(\frac{(-1)^n + 1}{1+n} + \frac{(-1)^n + 1}{1-n} \right) \\ &= \frac{(-1)^n + 1}{\pi} \left(\frac{1}{1+n} + \frac{1}{1-n} \right) = \frac{(-1)^n + 1}{\pi} \left(\frac{2}{1-n^2} \right) \end{aligned}$$

$$n \text{ is oneven} \Rightarrow a_n = 0$$

$$n \text{ is even } (n = 2k) \Rightarrow a_n = \frac{4}{\pi(1-n^2)}$$

$$\begin{aligned} \Rightarrow f(x) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) \\ &= \frac{2}{\pi} + \sum_{k=1}^{\infty} \frac{4}{\pi(1-(2k)^2)} \cos(2kx) \\ &= \frac{2}{\pi} - \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{1}{4k^2 - 1} \cos(2kx) \end{aligned}$$

References